

YUNHAI XIANG

May 20, 2026

PERSONAL DETAILS

Name Yunhai Xiang (in Chinese: 项云海, pronounced “yee-o-en-hi si-ang”)
Pronouns He/Him/His/Himself
Email yxiang72@uwo.ca (for encrypted emails: [PGP Public Key](#))
Website yunhaimath.com

EDUCATION

Western University 2025 — Now
Ph.D. in Mathematics (in progress)
Advisor: Chris Hall
Specialization: Number Theory, Arithmetic Geometry, AI for Math

Western University 2024
M.Sc. in Mathematics
Advisor: Chris Hall

University of Waterloo 2018 — 2023
B.Math. in Pure Mathematics, Honors, with Distinction

SKILLS

Colored by level of mastery: *Expert*, *Advanced*, *Proficient*, *Familiar*, *Beginner*

Languages Mandarin, English, French
Programming Python, Lean, C/C++, C#, Rust
Computing SageMath, NumPy, SciPy, Pandas, PyTorch, XGBoost
DSLs LaTeX, SQL
Tools Git, Cinema 4D, Blender, Unity, Godot, Bevy
Hobbies Magic Tricks

PUBLICATIONS

QEDBench: Quantifying the Alignment Gap in Automated Evaluation of University-Level Mathematical Proofs
Joint with Santiago Gonzalez, Alireza Amiri Bavandpour, Peter Ye, Edward Zhang, et al.
International Conference on Machine Learning (ICML) 2026

AWARDS

Mathematics Undergraduate Research Award 2022

- Highly competitive undergrad research award of value \$10000 at University of Waterloo. Worked on research project in summer related to Weyl groups, supervised by Prof. Matthew Satriano.

ACADEMIC MEETINGS & PRESENTATIONS

Conferences Attendance & Presentations

Attended academic conferences and given research or expository talks

- Using AI to Find Examples Workshop, under thematic program Universal Statistics in Number Theory, Winter 2026, Centre de Recherches Mathématiques, Université de Montréal
- University of Western Ontario Research in Computer Science (UWORCS), Winter 2026, UWO

- *Rigid Analytic Geometry*, Ontario Graduate Mathematics Conference, Summer 2024, UWaterloo
- *Cohomology of Grassmannians*, Canadian Undergraduate Mathematics Conference, Summer 2022, Université Laval

Seminar Attendance & Presentations

Attended academic seminars and given research or expository talks

- *Trace of Frobenius*, Winter 2026, Graduate Student Seminar, UWO
- *Brauer Groups of Fields Part II*, Winter 2026, Algebraic Geometry Student Seminar, UWO
- *Brauer Groups of Fields Part I*, Winter 2026, Algebraic Geometry Student Seminar, UWO
- *Weyl Group and Root Datum*, Summer 2025, Algebraic Groups Seminar, UWO
- *Hecke Operators*, Summer 2025, Algebraic Groups Seminar, UWO
- *Parabolic and Borel Subgroups*, Summer 2025, Algebraic Groups Seminar, UWO
- *Lie Algebras of Algebraic Groups*, Summer 2025, Algebraic Groups Seminar, UWO
- *Non-Archimedean Fields*, Summer 2024, Rigid Analytic Geometry Seminar, UWO
- *Admissible Topology*, Summer 2024, Rigid Analytic Geometry Seminar, UWO
- *Cones and Affine Toric Varieties*, Summer 2024, Toric Varieties Learning Seminar, UWO
- *Galois Cohomology*, Winter 2024, Graduate Student Seminar, UWO
- *Derived Categories*, Fall 2022, Algebraic Geometry Working Seminar, UWaterloo

TEACHING & MENTORING

Course Instructorship

- Coming soon.

Teaching Assistantship

Proctored and graded exams, graded homeworks, tutored at Help Centre, and taught tutorial sessions

- MATH 3150, *Elementary Number Theory I*, Fall 2024
- MATH 2155, *Mathematical Structures*, Fall 2025
- CALCULUS 1501, *Calculus II for Mathematical and Physical Sciences*, Winter 2025, Winter 2026, Summer 2026
- MATH 1229, *Methods of Matrix Algebra*, Summer 2024, Summer 2025
- MATH 1228, *Methods of Finite Mathematics*, Summer 2024, Summer 2025
- CALCULUS 1000, *Calculus I*, Winter 2024, Summer 2024, Winter 2025, Fall 2025, Winter 2026

Mathematical Outreach Lectures

Gave expository lectures to high school students or undergraduate students on advanced mathematical topics

- *Surreal Numbers and the Game of Go*, Summer 2025, Western Summer Math Camp (Grade 10)
- *Mathematical Magic Tricks: Gilbreath's Principle*, Summer 2024, Western Summer Math Camp (Grade 11)
- *Why is There no Quintic Formula*, Summer 2024, Western Summer Math Camp (Grade 10)
- *Category Theory Demystified*, Winter 2023, Short Attention Span Math Seminar (Undergraduate)

Directed Reading Program Mentorship

Mentored undergraduate students on reading projects, culminating in a presentation

- Jude McNorgan Lubos, *Introduction to Quadratic Reciprocity*, Winter 2026

SERVICE & VOLUNTEER WORK

Seminar Organization

Research or learning seminars organized or co-organized

- Algebraic Geometry Student Seminar, Summer 2026
- Algebraic Geometry Student Seminar, Winter 2026, co-organized with Alex Zwart
- Algebraic Geometry Student Seminar, Fall 2024, co-organized with Alex Zwart
- Toric Varieties Learning Seminar, Summer 2024, co-organized with Nikolas Rieke